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ORIGINAL RESEARCH

Transparency of combinatorial optimisations via machine learning and explainable AI

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Abstract
In this golden age of artificial intelligence, transparency and responsible decision-making are paramount. While machine learning (ML) and operational research (OR) optimisations are fundamental aspects of AI, the benefits of explainable AI (XAI) for combinatorial optimisations remain underexplored. This study investigates the convergence of XAI and OR, emphasising the importance of transparency in combinatorial optimisations. Using the Knapsack problem as an example, we demonstrate that interpretable ML models can effectively solve combinatorial optimisation challenges and enhance transparency. Additionally, we illustrate the application of post-hoc XAI methods to OR optimisations solved with ML, providing transparent, human-friendly explanations. The key contributions of this work include proposing the application of the SAGE framework for transparent OR, demonstrating the integration of XAI with combinatorial optimisations, and offering practical guidelines for creating transparent explanations. These contributions can aid decision-makers in understanding, communicating, and trusting combinatorial optimisation solutions, paving the way for enhanced transparency in operational research across various sectors.

Keywords Operational research · Combinatorial optimisations · Knapsack · Transparency · Interpretable machine learning · Explainable artificial intelligence

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1 Introduction

We are experiencing the golden age of artificial intelligence, with generative AI and self-driving cars appearing (Kaynak, 2021; Zhang & Lu, 2021). This growing adaptation of AI necessitates government bodies, health sectors, and businesses to understand AI's solutions and workings through transparent and responsible AI (Buiten, 2019). While AI is predominantly associated with machine learning, operational research optimisations (i.e. combinatorial optimisations) are a fundamental part of AI (Russell & Norvig, 2020; Garn, 2024). Explainable AI offering transparency and responsibility building on machine learning has been investigated extensively (Arrieta et al., 2020; Mill et al., 2023, 2024a). Yet, the link, applications and benefits of explainable AI in conjunction with operational research optimisations constitute a significant gap in the body of knowledge.

Machine learning (ML) is a data-driven approach based on learning algorithms such as decision trees, neural networks and support vector machines. Optimisations in Operational Research (OR) are often formulated via linear programs (LP), integer programs (IP) or mixed integer linear programs (MILP), which use exact approaches or metaheuristics to find solutions. Therefore, we distinguish operational research optimisation (aka combinatorial optimisation) from machine learning optimisation. Whilst machine learning optimisation (MLO) is mainly concerned with minimising a loss function for predictions or classifications, operational research optimisations (ORO) often have a real-world objective subject to constraints. This work investigates the synergies between the ML and OR domains and their implications for explainable artificial intelligence (XAI), which results in transparency.

Similar to the need for ML algorithms to be transparent, there is a growing demand for combinatorial optimisations to be more transparent. The transition from Industry 4.0 to 5.0 has placed a greater emphasis on human-centric AI, further motivating the need for transparency in AI and optimisation solutions (Turner & Garn, 2022). Turner and Garn (2022) emphasised the need to integrate explainable AI and utilise combinatorial optimisations in the next generation of discrete event simulators to provide transparency to decision-makers.

Incorporating transparency into analytical techniques used in operational research to improve interpretability, accountability, and the reliability of data-driven decision-making has been a relatively unknown territory to researchers and practitioners. While transparency in the context of machine learning has been explored (Arrieta et al., 2020), operational research (OR) literature lacks a definition of transparent OR. Some literature links OR to transparent machine learning but lacks tangible approaches in the context of operational research optimisations (Behl et al., 2023; Sabharwal et al., 2024). Turner and Garn (2022) show the need for more human-centric transparent discrete event simulators (DES) for manufacturing systems by suggesting a framework that includes XAI and combinatorial optimisations.

Combinatorial optimisations are applied in many practical applications such as public transport, environmental services, supply chain logistics and health care analytics. For instance, policymakers and decision-makers must justify why bus or train stations will or will not be serviced with a certain frequency or why a specific bus service is required (Ahern et al., 2022). The tool for their decision support may be based on underlying combinatorial optimisations, and the mathematical program's complexity and solution need a simple explanation. In this case, explanations can be on economic and sustainable grounds.

Another example is environmental services (Beltrami & Bodin, 1974; Datta & Kapoor, 2023), which are required to collect recyclable waste from households. The underlying combinatorial optimisation is a type of vehicle routing problem (VRP). Here, the environmental services must validate and justify to the authorities that the routes are economic and sustainable and consider societal needs. The solutions must consider aspects such as school zones and essential main roads being serviced at an appropriate time. Transparent operational research must demonstrate the fulfilment of these requirements.

Consider, for example, the growing use of optimisation algorithms in supply chain logistics. These algorithms decide on vehicle routing or warehouse stocking policies with real-world financial and operational impacts (Rohani et al., 2022). However, the lack of transparency in these decisions-particularly when they produce unexpected or counterintuitive results-can lead to confusion and reduced trust among human decision-makers. Suppose a logistics manager cannot understand why an algorithm chooses a specific route that, on the surface, seems suboptimal. In that case, they may be reluctant to implement the solution, even if it is mathematically optimal.

Resource allocation is another domain in the healthcare sector where transparency is vital (Das, 2022). Optimisation models allocate scarce medical resources, such as ICU beds or organ donations. Still, decision-makers must be confident that these solutions optimise efficiency and adhere to ethical and legal standards. Without transparency, stakeholders may question the fairness of the algorithm's decisions, particularly in high-stakes environments where human lives are affected. In these cases, explainable AI can illuminate the decision-making process, offering insights into how and why certain solutions were reached.

The real-world scenarios above highlight the importance of transparent and explainable operational research optimisations. Although recent studies in the health (Bastani et al., 2022) and property sector (Nyawa et al., 2023) show transparent and explainable methods in an operational research context, they miss the link to combinatorial optimisations.

In effect, while transparency in machine learning has been explored extensively (Arrieta et al., 2020), there remains a significant lack of tangible and general methodologies that integrate explainable AI (XAI) with combinatorial optimisations, particularly in operational research contexts. This paper addresses this gap by introducing novel applications of the SAGE framework to create transparent, interpretable solutions for combinatorial problems. Unlike prior works that offer transparency in machine learning but not for OR-specific challenges (Molnar, 2023a, b; Schmidt et al., 2020), our approach provides practical insights for real-world OR problems. Further, by incorporating post-hoc XAI methods in novel ways and adapting them to combinatorial optimisation, we aim to push the boundaries of what XAI can achieve in OR contexts.

Table 1 summarises the above articles regarding focus area and transparency.

In sum, this paper addresses several knowledge gaps: (1) the lack of a clear link between interpretable ML and combinatorial optimisations with a focus on transparency, (2) the limited application of post-hoc XAI techniques to combinatorial optimisation solutions derived from ML models, and (3) the scarcity of transparent, human-friendly explanations for complex OR optimisation problems. We aim to bridge these gaps by providing concrete approaches that offer transparency to combinatorial optimisations using machine learning and explainable AI.

This work aims to apply machine learning and explainable AI to operational research optimisations to gain transparency. In this respect, the key objectives are to (i) Contextual-

Table 1 Comparison of our work with existing literature

References	Focus area (ML, OR, CO)	Transparency method	Key contribution to transparency in OR
Ahern et al. (2022)	Bus services, OR, CO	Result based (Pareto)	Pareto front solutions for decision-makers but do not cover XAI and ML
Arrieta et al. (2020)	ML	Post-hoc XAI	Focuses on machine learning transparency but lacks OR-specific applications
Bastani et al. (2022)	Health-care, ML	Empirical Bayes	Demonstrates the need for trust and transparency for policymakers but lacks context to CO
Behl et al. (2023)	Supply Chain, OR, ML	High-level responsible AI & XAI needs-analysis	Emphasises fairness and accountability in supply chain performance, not focusing on tangible CO transparency but on organisational needs and insights
Rohani et al. (2022)	Logistics, OR, CO	Visualisations (VR)	Embed trust by allowing managers to interact with the CO analysis tool, but lacks ML and XAI
Sabharwal et al. (2024)	ML	XAI (LIME, SHAP)	Suggests integration of XAI with OR but does not offer a framework for CO transparency
Turner and Garn (2022)	OR (DES, CO)	XAI in DES framework	High-level framework for next-generation DES integrating CO and XAI
Our work	OR, CO, ML	SAGE Framework with Post-hoc XAI	Introduces a general methodology for combinatorial optimisation transparency with human-readable explanations

ise transparency for OR applications using the SAGE framework (Mill et al., 2024a), (ii) Demonstrate that combinatorial optimisations can be solved using interpretable machine learning methods, (iii) Show and apply relevant XAI methods in the context of optimisations, and (iv) Offer human-friendly explanations to enhance transparency.

Hence, the key contributions of this paper are as follows:

- Demonstrating how interpretable machine learning can be applied to combinatorial optimisations;
- Proposing the use of a general framework (SAGE) to gain transparency;
- Illustrating the usage of explainable AI to combinatorial optimisations and
- Providing pointers to create transparent human-readable explanations.

The rest of this paper is organised as follows. Section 2 reviews literature about explainable AI and transparency to provide foundations and highlight potential extensions related to the paper's aim. SAGE is especially reviewed as the framework used in this work, which gives XAI the transparency dimension. Furthermore, it introduces the Knapsack problem and its applications, as this will be the guiding example throughout the paper. In Sect. 3, we begin by contextualising transparency to show the importance of the intended audience. This section applies the SAGE framework and offers a convenient way to structure setting, audience, goal and ethics, supporting the creation of transparent, human-readable explanations later. To benefit from XAI methods, and out of general importance, Sect. 4 investigates the application of an interpretable machine learning approach to operational research optimisations, highlighting synergies between these two areas. Explainable AI is then investigated in the context of combinatorial optimisation in Sect. 5. Finally, Sect. 5.2 gives pointers to create transparent human-readable explanations, and Sect. 6 provides some concluding remarks.

2 Related work

Artificial intelligence (AI) is the ability to think and act like humans or more rationally. We proposed this definition in Turner and Garn (2022, s5) which was derived from the definitions of the AI term in Russell and Norvig (2020) and Buiten (2019). The advantage of this definition is its compactness. Yet, to understand what is behind AI, it is more appropriate to consider the typical topics covered in the AI literature. In Garn (2024, s9.1) we gave an overview of these areas and showed that machine learning and combinatorial optimisations are essential topics in the AI domain. Yet, topics such as explainable and responsible AI are still fairly recent. In this section, we briefly introduce the interrelated concepts of responsible AI, responsible analytics, and explainable AI and how these concepts relate to or act as building blocks for better transparency. This work aims to bridge the generic transparent AI domain and combinatorial optimisations using the knapsack problem as an example. Hence, we review the knapsack problem (amongst other OR optimisations) and its applications with a focus on ML approaches (e.g. neural networks) as solution methods.

2.1 Explainable AI

Explainable AI is a research field that aims to make AI systems' results more understandable to humans. Thus, it increases the transparency of the system. An explainable AI method provides a "convincing" argument (explanation) to the audience about the result, encouraging trust in the underlying method. Sometimes, explainable AI provides a narrative or indication of major algorithm steps, increasing transparency (Ehsan et al., 2021). Their work encourages social transparency using scenarios, a view we share in this work. Gilpin et al. (2018) treat XAI as an umbrella term to summarise reasons for neural network behaviour, gain users' trust or produce insights about the causes of their decisions. Gunning and Aha (2019) see XAI as "*AI systems that can explain their rationale to a human user...*". The understanding of what explainable should constitute is still divided, with some still focusing on experts as the audience. Back in 2017, Miller's survey found that XAI was primarily for researchers rather than the audience (Miller et al., 2017). Later, he provided several interesting insights from social sciences supporting the development of the XAI domain (Miller, 2019). This

work allows researchers from "traditional" AI to access the social science (psychology, cognitive sciences, philosophy,...) knowledge base of explanations. For instance, for an AI to produce better explanations, one can make use of the paradigms of explaining, such as (1) explaining by contrasting events, (2) explaining by selecting only a few (two or three) facts, (3) use causal explanations rather than probabilistic arguments, and (4) tailor explanations according to the audience. Now, explanations have found their way to a far wider audience than researchers, such as the gaming industry. For instance, human-friendly explanations for chess moves are provided in chess.com and can be used for Stockfish (Hertl, 2022). It provides human-friendly analysis explanations such as *"The pawn is adequately defended. This is the only good move!..."* (Garn, 2023). We observe that it makes use of the explaining paradigms mentioned above. In Sect. 5.2, we discuss human-frindly explanations in more detail. Fraud detection is often found as an XAI research case study, but there is limited evidence that it is being used by the industry (Mill et al., 2023). To encourage the uptake of XAI in this industry, Mill et al. (2023) research agenda encourages to (1) give explanations within a specific context, (2) increase focus on intrinsic models, (3) measure the faithfulness of explanations, and (4) investigate human interaction with explanations. However, as we see in this work, the agenda also applies to the operations XAI industry. Other literature has listed many more XAI applications in the areas of business, manufacturing, energy and others (Ahmed et al., 2022; Le et al., 2023), which benefit from the agenda. Section 3.1, shows the SAGE framework that integrates aspects such as context, audience, paradigms of explanations and human-friendliness. Systems and methods capable of integrating this will be called *second-generation XAI methods*.

Considering the current taxonomy of XAI methods, we can observe a good starting point for the mentioned agenda. We will call them the *first-generation XAI methods*.

Figure 1 shows the taxonomy of XAI methods; a more refined view can be found in Arrieta et al (2020, Fig. 6). Our work will primarily deal with specific local methods. Generally, XAI methods are classified via their model type or scope. Most research focuses on model-agnostic methods, meaning the XAI method is independent of the underlying ML model. In this approach, an AI method learns a model, which can be treated as a black box, after which a post-hoc XAI method derives an explainable model. In contrast, instead of being model-agnostic, the XAI method can be model-specific. These methods, also known as intrinsic methods, are inherently interpretable from the outset. In Sects. 4.3 and 4.4, we use decision trees to interpret feasible solutions, a prime example of intrinsic XAI. The advantage of intrinsic XAI is that the model can be built with an explanation in mind. In the XAI literature, intrinsic XAI is relatively less considered than post-hoc XAI. Models resulting from intrinsic XAI are known as *transparent models*. Table 2 provides a short list of popular intrinsic XAI methods.

Typically, intrinsic XAI sacrifices accuracy in favour of interpretability.

Fig. 1 XAI taxonomy of methods

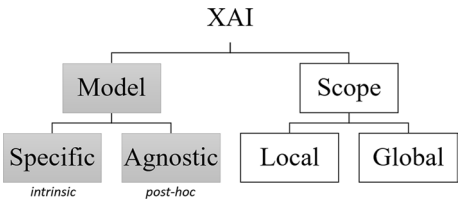

Table 2 Intrinsic XAI methods: overview and applications for interpretability and transparency in decision-making

Abbr	Name	Description	Applications	Strengths
DT	Decision tree	A tree-structured model that splits data into decision nodes to arrive at a target prediction	Business, healthcare, customer analytics	Highly interpretable, intuitive visual representation of decisions
LM	Linear model	Models the relationship between a dependent variable and one or more independent variables using linear equations	Finance, economics, operations research	Easy to interpret, widely used in regression analysis
GAM	Generalized additive model	Extends linear models by allowing non-linear relationships between dependent and independent variables using smoothing terms	Energy, environment, medicine	Balances flexibility and interpretability, suitable for non-linear patterns
kNN	k-nearest neighbors	A non-parametric method that classifies or predicts by looking at the closest k neighbours to a point	Image recognition, stock prediction, fraud detection	Simple, highly interpretable for low-dimensional data, effective for small datasets
LogR	Logistic regression	Predicts binary outcomes based on a logistic function, often used for classification tasks	Medical diagnosis, marketing, credit scoring	High interpretability in binary classification tasks, easy to implement
BM	Bayesian models	utilizes bayesian inference to estimate parameters based on prior distributions and observed data	Decision-making, weather forecasting, risk analysis	Offers uncertainty estimates and robust decision-making in the presence of uncertainty

The scope of the XAI may be global or local. A global XAI model gives an explanation for the entire population (all instances). Table 3 gives a list of methods and examples.

Local XAI models explain a single instance, which should be stable in its neighbourhood. Table 4 gives a few examples of local model-agnostic methods.

LIME and SHAP are the most popular local-model agnostic methods (Mill et al., 2024b, Table 2). Hence, we discuss them in more detail and use LIME Sect. 5.1 to explain the feasibility and optimality of combinatorial optimisation solutions.

Local interpretable model-agnostic explanations (LIME) were first introduced by (Ribeiro et al., 2016). LIME finds an explanation for an instance by generating several cases around it using a given model (e.g. random forest). The explanation is typically in the form of an interpretable model (e.g. *k*-LASSO). Typically, these models reduced the model's complexity, especially regarding the number of features used. However, approaches and implementations dealing with creating features and aggregating the feature's domain are sparse, with Biecek (2018) providing ideas. Section 5.2 addresses this shortcoming. In the context of combinatorial optimisations, we use LIME in Sect. 5.1 to explain an instance's feasibility and to interpret the solution's optimality.

In 1951, Lloyd Shapley introduced a solution concept for cooperative game theory. This concept is based on Shapley values, which have been successfully applied in the machine learning domain to explain predictions. It determines the contribution of all features fairly for a prediction, while LIME does not guarantee this. The Shapley values allow contrasting explanations by changing the comparison from average to a single data point (Molnar, 2023a, b). Sabharwal et al. (2024) discuss the benefits of Shapley additive explanations (SHAP), which provide finance managers with explanations for detected anomalies. They follow the classic machine-learning roadmap similar to Garn (2024, s5.3) and provide SHAP XAI visualisations. This demonstrates the power of using XAI methods to obtain local explanations. The work suggests the context of transparency, AI, XAI and OR in a general manner, and stops at the well-established first-generation XAI methodology suitable for an expert audience.

A comprehensive review of tabular XAI methods can be found in Sahakyan et al. (2021). Molnar (2023a) provides an introduction to some of the XAI methods. Overall, we can see that we are entering the area of second-generation XAI methods, whereas first-generation methods such as LIME and SHAP are the basis for offering the next level of human-friendly explanations. The call for responsible AI is becoming an integrated part of the second-generation XAI.

2.2 Responsible AI

Typically, the previously mentioned XAI methods use features without considering whether they are responsible from an ethical or societal perspective. This has triggered regulators and policymakers to demand them and introduce more transparency (Buiten, 2019). Transparency is essential in sectors such as health and social care, criminal justice and financial services (Centre for Data, Ethics and Innovation, 2020). Arrieta et al. (2020) specify *responsible artificial intelligence* as a methodology for the large-scale implementation of AI methods in real organisations with fairness, model explainability and accountability at its core. Similarly, Mikalef et al. (2022) argue that safer and more effective technologies can be developed by considering responsible AI principles such as algorithmic transparency.

Table 3 Global model-agnostic methods: overview and applications for overall model behaviour based on entire dataset

Abbr	Name	Description	Applications	Strengths
PDP	Partial dependence plot	Plots the relationship between the target and one or more features while averaging out other features	Credit scoring, marketing	Offers visual insights into the influence of features across a global population
ALE	Accumulated local effects	Like PDP but better suited for correlated features, showing the marginal effects of features	Predictive maintenance, healthcare	Handles feature dependencies better than PDP, making results more reliable
FIRM	Feature importance ranking measure	Ranks the importance of features based on their impact on model performance or prediction accuracy	Fraud detection, customer analytics	Provides clear and interpretable rankings of feature contributions
SP-LIME	Submodular pick LIME	An extension of LIME that selects a set of explanations to explain multiple instances in a dataset	Text classification, medical imaging	Covers multiple instances with diverse explanations, increases model generalisability
FD	Functional decomposition	Breaks down complex functions into simpler components to understand how individual elements contribute to the whole	Energy, signal processing	Allows for fine-grained understanding of complex systems and processes

Table 4 Local model-agnostic methods: overview and applications for explaining individual predictions and end-users looking for actionable insights on specific instances

Abbr	Name	Description	Applications	Strengths
LIME	Local interpretable model-agnostic explanations	Explains individual predictions by approximating the black-box model locally with interpretable models	Text classification, image analysis	Simple, effective for understanding local decisions, works for any model type
SHAP	SHapley additive exPlanations	Provides fair, additive feature importance values by allocating contributions fairly across features	Finance, fraud detection	Consistent and theoretically grounded, ensures fairness in feature contributions
Anchors	Scoped rules	Uses high-precision rules to explain local instances by showing the parts of input that are critical for prediction	Healthcare, law enforcement	High precision in identifying key features in local predictions, robust to changes
CF	Counterfactual explanations	Provides examples of how a feature's change would alter the prediction, enhancing interpretability	Loan approval, insurance claims	Helps users understand causality, showing how small changes impact predictions

Further, Behl et al. (2023) investigate the role of responsible AI in improving the performance of supply chains. They received responses from 414 decision-makers who deploy AI in supply chains, which allowed them to link responsible AI to fairness, accountability, sustainability and transparency. Furthermore, a line to supply chain performance was derived, where the term performance is generic.

De Bock et al. (2023) define *responsible analytics* as the development or improvement of advanced methods for transforming data into insights in pursuit of compliance with societal expectations, such as ethical, legal, or frugal norms. Responsible AI or Responsible Analytics complements or extends current XAI by adding aspects such as fairness, ethics, and legal considerations. This work uses the SAGE framework (Sect. 3.1), which covers Responsible AI using the "Ethics" dimension.

A few studies have provided frameworks and roadmaps for integrating XAI with OR. For instance, De Bock et al. (2023) introduced a framework based on machine learning and XAI for some operational research challenges. Their work summarised machine learning and XAI methods. It suggested considering performance analytics, responsible analytics and attributable analytics. These three domains can be associated with prescriptive analytics, transparent AI and explainable AI. Their proposed framework covers most of the classic machine-learning roadmap (Garn, 2024, s5.3). Additionally, they offer an experimental design stage, an interpretation and an evaluation stage. Unfortunately, the provided OR examples are too generic to explain how transparent and explainable AI output can be achieved for combinatorial optimisations.

2.3 Transparency

Approaches to explain AI can be divided into those offering more transparency about the AI model's outcomes and AI's methods. Guidotti et al. (2018) offer the following four "opening the black box" types: (1) an explanation of the model, (2) an explanation of the outcome, (3) an inspection of the model, and (4) designing a transparent model. This constitutes a categorisation of explanations as argued in Mill et al. (2023). Lipton (2018) considers transparency with respect to machine learning's model workings. Here, he distinguishes the model's entirety (which they call simulatability), its components (decomposability), and its training algorithm. He suggests that transparency is a form of inseparability in textual or visual form. This is associated with *algorithmic transparency*. Generally, a goal of transparent AI is to increase trust in the system, but it has been shown that it can lead to the opposite (Schmidt et al., 2020).

The audience, such as stakeholders, developers, and users affected by AI decisions, is essential when designing and deploying transparent AI systems (Felzmann et al., 2020). Here, it is necessary to ensure the processes, decisions, and actions are clear and comprehensible to the audience to ensure trust, accountability, and ethical use. Felzmann et al. (2020) suggest using the principle of transparency by design, which provides implementable measures.

Visualisation techniques can provide transparency for black box models such as neural networks, support vector machines and random forests using sensitivity analysis (Cortez & Embrechts, 2013).

Solutions of optimisation models for problems concerning the shortest path, travelling salesman, and vehicle routing can be visualised as graphs or in geographical information systems (Garn, 2021, 2020). Familiar to most is the implementation of shortest path algorithms within SatNavs. Here, transparency is provided by seeing the type of road used, congestion and alternative routes. Solutions for transportation and trans-shipment models can also be displayed efficiently (Garn, 2020). Generally, the OR literature displays the problems and solutions of optimisation challenges through graphs. For instance, Garn (2021) visualises forklifts in a warehouse and maps it to the balanced travelling salesmen problem (TSP) to increase algorithmic and conceptual transparency.

2.4 Machine learning solves combinatorial optimisations

In 1999, Smith reviewed a decade of research about neural networks for combinatorial optimisations (Smith, 1999). The TSP was the focus point, and numerous successful approaches were used here. Applications of neural networks for solving assignments, constraint satisfaction, cutting stock and packing problems were cited. The practical importance of the Knapsack problem and its ML literature were discussed. More than 20 years later, Bengio et al. (2021) reviewed the usage of machine learning to support combinatorial optimisations. They themed the ML techniques by (1) end-to-end learning, (2) learning-to-configure and (3) ML alongside optimisation. Here, end-to-end learning is defined as learning a machine learning model that provides a solution for a given optimisation problem instance. Learning-to-configure provides additional information for a combinatorial optimisation algorithm to adjust its performance. Running machine learning alongside optimisation algorithms allows the repeated application of the learning-to-configure ML model. For our work's aim, end-to-end learning is most appropriate, also recognising that the themes (2) and (3) could be used in conjunction with (1). Vinyals et al. (2015) propose a pointer network based on recurrent neural networks to solve operations research optimisations. They demonstrated that solutions for travelling salesman problems (TSP), convex hulls and Delaunay triangulations can be found. However, existing TSP algorithms were used to learn the machine-learning models. A recent work, published during this paper's revisions, proposes XAI and a large language model (LLM) for the vehicle routing problem (VRP) (Kikuta et al., 2024). Additionally, they discuss counterfactual explanations.

2.5 Knapsack problem

The Knapsack problem is a classic problem in OR. Given a set of items, each with an associated weight and value, and a knapsack with a weight capacity limit, the objective is to select a combination of items to 'pack' in the knapsack such that the (i) total weight does not exceed the capacity and (ii) the total value is maximised.

The knapsack problem has a wide range of real-life applications, from resource allocation in manufacturing to portfolio optimisation and investment management in finance (Salkin & De Kluyver, 1975; Assi & Haraty, 2018). Additionally, the knapsack problem can be utilised in online advertising to optimise selecting ads to display on a webpage, considering factors such as ad revenue, relevance, and available ad space (Langendoen et al., 2024). In logistics and transportation, the knapsack problem can be used to optimise the selection of items to be transported in a vehicle with limited carrying capacity, considering factors like weight, volume, or profit (Piedra de la Cuadra, 2023). Similarly, in shipping and logistics, the knapsack problem can be used in Container Loading to efficiently load containers with various goods to utilise space optimally and minimise shipping costs (Chien et al., 2009). Lee and Hsu (1989) are among the first to solve the multi-constraint knapsack problem using neural networks. They demonstrated a faster runtime of the neural network model than the optimal solution found by a branch-and-bound algorithm. Other early work using neural networks focused on the mathematical aspects of solving the knapsack problem (Aourid et al., 1995) where Ohlsson et al. (1993) solved problems of up to 10,000 items. The problem continues to be analysed and solved with neural networks (Hertrich & Skutella, 2023). Recently, some work has been done in the domain of XAI on counterfactual explanations for the knapsack

problem (Korikov & Beck, 2021; Korikov et al., 2021; Segal et al., 2023). A counterfactual explanation originates from a contrastive question such as "Why was the decision A and not B?" where the differences in the underlying reasons are used. A counterfactual explanation provides "what-if" information in terms of which alterations of the input features could change the output of a predictive model. A counterfactual explanation is then defined as the smallest change to the "world" (as captured by the input data) that can be made to obtain a desired outcome (Wachter et al., 2017; Confalonieri et al., 2020).

It can be summarised that the current research about responsible AI, in conjunction with operations research, captures the well-established domains of AI and explainable AI and provides an extension into social sciences. To date, the OR applications of transparent and explainable AI implementations appear to be too abstract to be used or too similar to computer science challenges. This indicates we have not been able to find implementations that generate actual transparent or responsible AI solutions that associate the above to classic Operational Research optimisation challenges such as the MILP.

3 Transparency for OR via SAGE

Based on the analysed literature in Sect. 2, the following four approaches can be used and applied to increase the transparency of optimisation solutions. First, a *visualisation* of the problem and its solution can be provided. This can include a graph or chart showing the problem's components and their interaction. Second, the *logic* behind the solution is explained. This includes explaining and elaborating on the assumptions and calculations to arrive at the solution. Third, a detailed explanation of the *constraints and objectives* used to formulate the problem is provided. Fourth, a *step-by-step* approach to explaining the solution is used. This could include a flowchart or a series of diagrams showing the different steps to arrive at the solution. Finally, the *audience* is involved in the process and is invited to provide feedback and input on the solution. This will help to ensure that the solution is transparent and meets the needs of all involved. While these actions are a viable starting point for explaining optimisation solutions, they lack the theoretical support and structure required for the XAI discipline. Hence, we apply the SAGE framework to the Knapsack challenge as preparation for creating an XAI that has the audience as its centre. We then introduce the Knapsack problem and demonstrate how to increase its operational research transparency without needing XAI.

3.1 SAGE framework

The SAGE framework was introduced initially in Mill et al. (2024a), which serves as a design aid for researchers and practitioners in the application-specific development of XAI. Figure 2 provides an overview of the framework.

We briefly define the most important terms before applying them to our specific XAI context. The audience is defined as the immediate recipient of the system-delivered explanation. As noted in Mill et al. (2024a), two roles interact mainly with the explanation: the creator, i.e., the business analyst or researcher proposing a solution strategy, and the operator, who applies and uses the proposed strategy and directly interacts with the system inputs and outputs (Mill et al., 2024a). Goals satisfy the questions of a specific audience, and Set-

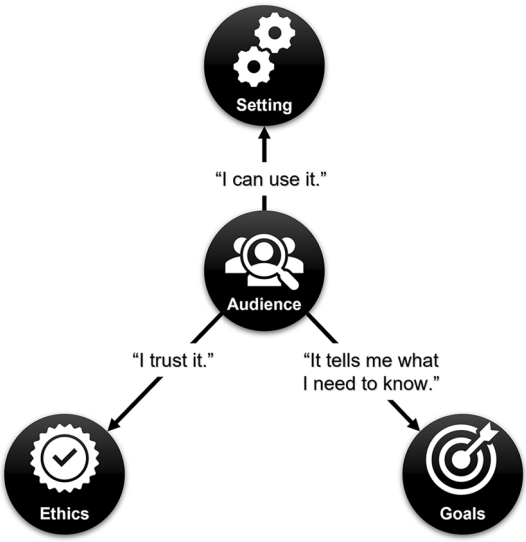

Fig. 2 The SAGE framework - a design aid for researchers and practitioners in the application-specific development of XAI. [Source: Mill et al. (2024a)]

ting refers to factors that affect the usability and comprehension of the explanation. Finally, *Ethics* is based on truthfulness and fairness. The SAGE framework was derived by considering the context identified from various views in the literature. In Mill et al. (2024b), the real-world efficacy of explainable artificial intelligence using the SAGE framework and Scenario-Based Design (SBD) is demonstrated, which is well aligned with the work presented here. However, we use the framework in an OR optimisation context in this work.

The framework is kept general and open to accommodate different authors' viewpoints. Generally, the view is that explanations are contextual (Miller et al., 2017). A scenario-based example of contextualising the SAGE terms in a production setting is as follows:

- Audience—"Who I am": I am the logistics manager.
- Goals—"It tells me what I need to know": I want to know why this is a feasible allocation, why this is an optimal selection of items to be packed, and why these items were packed. I am obtaining a textual and visual answer.
- Setting—"I can use it": The model is used in industry. The combinatorial model is accurate. The interpretable model and post-hoc AI are reliable. The explanations should be text-based and human-friendly. They should be based on the system's components.
- Ethics—"I trust it": The provided arguments are conclusive and transparent. The sustainability and responsibility arguments are convincing. The explanations correspond to the workings of the model.

Generally, SBD is well established in human interactions (Carroll, 1997) and operations (Freeman et al., 2016), but its link to XAI is scarce, with only a handful of publications available at the time of writing. Wolf (2019) gives a case study demonstrating that the linkage leads to more effective solutions for the audience. Sun et al. (2022) apply SBD with XAI in the context of generative AI for code.

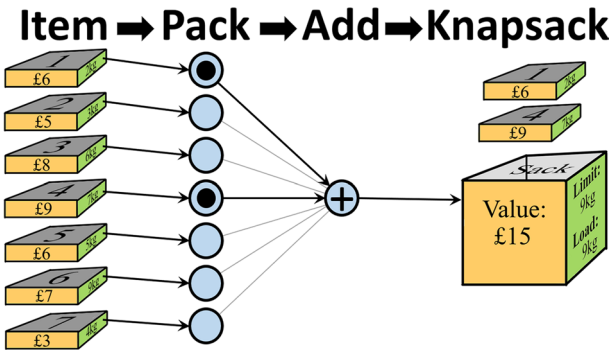

Fig. 3 Visualisation of components, their interaction and the solution's logic

Before we discuss the above XAI guiding principles and link the SBD to XAI, we will consider the general OR transparency approaches introduced at the beginning of this section for the knapsack problem.

3.2 Knapsack

The Knapsack problem is a well-known problem in Operational Research as a combinatorial optimisation challenge. It will be used as an example to support the previous four statements about increasing transparency. We mentioned that visualisations, logic, constraints & objectives, step-by-step approach, and audience increase transparency. The objective of the knapsack challenge is to pack items to maximise the value of the knapsack.

We begin by introducing the visualisations and logic for the Knapsack example (Fig. 3). The visualisation shows the components. The physical components are the items and the sack (including their identifiers), with their related contextual components, the weights and values. Furthermore, we illustrate the components' interactions with each other, such as whether to pack an item, the adding activity, and the knapsack's weight limit check. These interactions demonstrate the solution's conceptual logic. This can be identified with the setting within the SAGE framework.

Formulating the constraints and objectives improves the conceptual logic.

In operational research, a mathematical program specifies the problem formulation. A formal definition for the Knapsack problem, in the form of a binary program, is

$$\begin{aligned} \max_{x \in \mathbb{B}^n} \quad & \sum_{i=1}^{n} v_i x_i \\ \text{subject to} \quad & \\ & \sum_{i=1}^{n} w_i x_i \leq \hat{w} \end{aligned} \tag{1}$$

where v_i represents the value of the item i, w_i the corresponding weight[1] and $\hat{w}$ the weight limit. The decision variable x_i is one when an item is packed and zero otherwise.

[1] Other authors, e.g. Wolsey and Nemhauser (1999) use cost instead of weight.

An operational researcher may be content with the mathematical problem formulation. An industry analyst, however, may prefer to see the actual data and formula in a familiar tool such as Excel. Table 5 shows a Knapsack instance. This can be associated with the goal dimension within the SAGE framework.

Additionally, addressing the targeted audience increases transparency, which is the centrepiece of the SAGE framework. When stating the objective and constraints, we introduce common language terms such as item, pack, weight and value. While a worker may only want to know which items need to be packed, a statement such as "pack items one and four" may suffice. A manager's information must be communicated with the worker, the industry analyst, and other stakeholders. An informal introduction to this challenge offers transparency aspects, which can be used to motivate human-readable concepts. These concepts can be linked to traditional XAI methods. Hence, we must offer a story rather than merely a formal definition. The SAGE framework provides such a conceptualisation. In Mill et al. (2024a), the SAGE framework was proposed with the audience as the centrepiece transcending to transparent OR and XAI. Sections 4 and 5 return to the classic machine learning and XAI domain but demonstrate the linkage to combinatorial optimisations with overall guidance by the SAGE framework.

4 Intrinsic XAI for combinatorial optimisations

There is a well-established body of knowledge about approaching supervised learning challenges and its extension to XAI. Our literature review indicates that there is a lack of interpretable machine learning to offer transparency for combinatorial optimisations. It is interesting to note that black-box models specifically adapted to combinatorial optimisations have been the preferred approach in the reviewed literature in Sect. 2. However, as our key methodological contribution, we propose an approach based on the well-established machine learning roadmap (Garn, 2024, s5.3) to derive one associated with solving combinatorial optimisations via ML.

In this section, we demonstrate the details of our OR-ML methodology (as a step-wise procedure) using a Knapsack example and discuss its theoretical and practical implications. The learning algorithm generates an interpretable model, allowing us to benefit from intrinsic XAI and link it to the SAGE framework. When accuracy measures are of greater importance than intrinsic explainability, a black-box algorithm (see Sect. 2.1) should be used instead, followed by the XAI approach introduced in Sect. 5.

4.1 OR-ML methodology

We propose the following procedure, serving as a practical guideline:

Table 5 Knapsack instance in Excel with problem and solution visualisation

	max
Total Value	**15**
Packed weight	9
Weight limit	≤9

Item	*1*	*2*	*3*	*4*	*5*	*6*	*7*
Pack	b 1	b 0	b 0	b 1	b 0	b 0	b 0
Weight	2	3	6	7	5	9	4
Value	6	5	8	9	6	7	3

1. Formulate optimisation challenge (e.g. Knapsack, Dynamic Lot Sizing, etc.),
2. Generate a dataset of random solutions (feasible and infeasible),
3. Determine feasibility,
4. Learn to predict (classify) feasibility (e.g. Classification tree),
5. Synthetic Oversampling of Minority Class (assert feasibility),
6. Compute solution values for "feasible" solutions,
7. Learn to predict (regress) solution value (on the feasible data set), and,
8. Compare with the actual optimum.

The details of this methodology are given in the subsequent subsections.

4.2 Formulation and random solutions

Optimisation challenge: At this step, we formulate the OR problem. In Sect. 3.2, we provided the Linear Program 1 defining the optimisation challenge. Additionally, we must specify the optimisation instance and provide the underlying data. Here, we will use a small dataset to demonstrate the principle better. Assume that $n = 7$ items are available. The corresponding weights and values are the same as in Fig. 5. That means the weights are:

$$w' = (2, 3, 6, 7, 5, 9, 4). \tag{2}$$

The weight limit is $\hat{w} = 9$. Hence, all items qualify to be packed. The corresponding values are:

$$v' = (6, 5, 8, 9, 6, 7, 3). \tag{3}$$

As can be seen, the item weights and values do not follow any particular order.

Furthermore, the weights determine the solution's feasibility. To have meaningful feasibility ranges, it appears to be natural to order the items by weight. This allows cutting off heavy features (= items = elements) and identifying small gap-filling elements. The newly ordered weight and value vectors are:

$$w = [2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 9] \tag{4}$$

and

$$v = [6 \quad 5 \quad 3 \quad 6 \quad 8 \quad 9 \quad 7]\,. \tag{5}$$

The associated decision variables will adapt to this new order. For instance, item two has weight $w_2 = 3$, value $v_2 = 5$ and decision variable x_2. However, when learning how to predict the solution value, ordering by value or benefit is appropriate– the *benefit*b_i of item *i* is defined as its value-to-weight ratio $b_i = \frac{v_i}{w_i}$.

Random solutions: ML algorithms are data-driven. Hence, we must generate sample data to learn whether a solution is feasible. In our initial Knapsack example, we have got $n = 7$ items. That means the solution space consists of 2^n elements. In other words, there are 128 ways to pack seven items without considering a weight limit. We will generate a random

sample $X = (x_{ij})$ of $n_s = 100$ solutions without repetition, where *i* is the row (aka. record or observation) and *j* is the column (aka. feature or decision variable). Here is an illustration of the first six rows of *X*.

$$X_{1:6,:} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$

Typically, in machine learning, we denote a column *j* of *X* as x_j. This could be confused with the naming of the decision variable. However, we hope this fine distinction will be clear from the context. That means, in the OR domain $x_i \in \mathbb{B}$ and in the ML domain $x_j \in \mathbb{B}^{n_s}$. A technical remark on the generation of *X* is as follows. First, we generated n_s random integers between 0 and $2^n - 1$ without repetition, and then each integer was converted into a binary number. This triggers the idea of using integers in the first place for optimisations and machine learning. However, we will leave this idea for future research.

Generally, the sample size for a machine learning challenge is an interesting topic. The interested reader is referred to Hastie et al. (2009). Here are a few technical pointers. Memory can be an issue. For instance, 30 decision variables can exhaust 1GB (1,073,741,824B) of RAM (however, the presence of infeasible solutions reduces the space). A workaround is to consider only parts of the solution space. The second issue is time and associated CPU speed. Depending on the application, results must be found in seconds or hours. An alternative to the sampling approach could be a canonical selection of features or a systematic combinatorial feature generation.

4.3 Feasibility space

Determine feasibility: The sample can be divided into feasible and infeasible solutions using the program's constraints.

In the Knapsack example, the weights determine the feasibility of the solution. A solution *i* is feasible when

$$\sum_{j=1}^{n} w_{ij} x_{ij} \leq \hat{w}. \tag{7}$$

We denote the feasibility of a solution with y_i a 0-1 variable. A statistic of unexplored features may be good: a feature x_j is *unexplored* when all its elements are zero, i.e. $\sum_{i=1}^{n_s} x_{ij} = 0$. Insights that can be gained by considering the feasible solutions are:

- Likeliness of features (decision variables) being present in a feasible solution;
- Maximum number of features present in a feasible solution and
- Average number of features in a feasible solution.

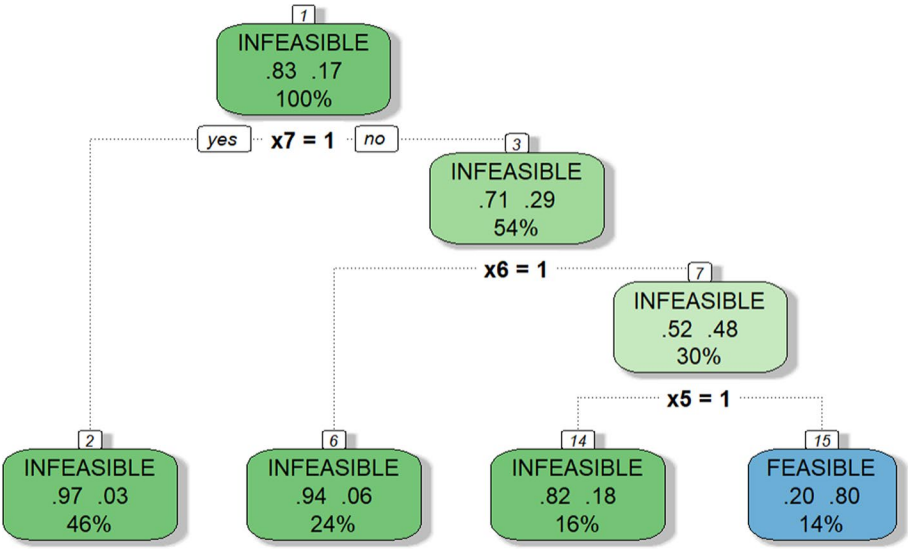

Fig. 4 Classification tree for feasible solutions

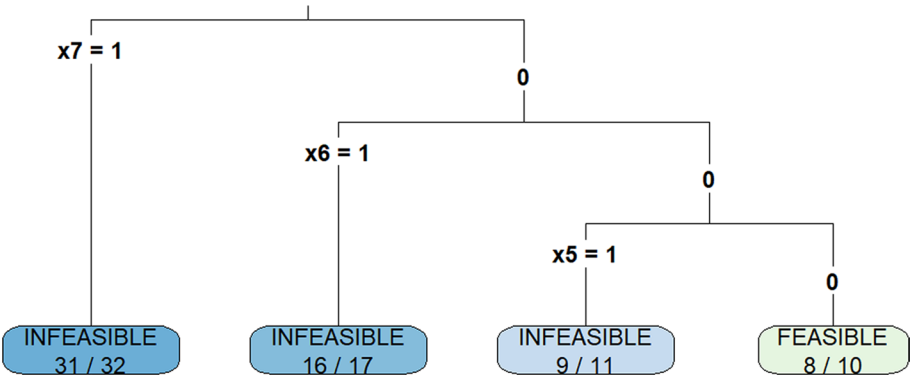

Fig. 5 Feasibility tree stating the decisions and number of solutions

These insights increase the transparency of solutions. Furthermore, they can be used to provide human-friendly explanations. For instance, if the audience requires an explanation of a solution's infeasibility, a statement like "*too many items were packed*" could be offered.

Learn feasibility: At this stage, we have derived X and y, and can learn whether a solution is likely feasible or infeasible. This is important when the solution space is too large. What are the benefits of knowing the likely characteristics of feasible solutions? Later, these insights can be used when we want to provide human-friendly explanations.

Classification trees are famed for their interpretability. For the Knapsack example, the classification tree is shown in Fig. 4.

As is common in ML, the data was split into training and test data. The tree was learned using 70% of the training data. A standard classification tree algorithm (Therneau & Atkin-

son, 2023) with its default control parameters was used to learn the classification tree model. Let us quickly interpret the tree. The root node tells us that 83% of the observations are infeasible, i.e. only 17% have feasible solutions. The decision to pack item 7 leads to 46% infeasible solutions (without any feasible solutions in the sample).

On the other hand, if we do not pack this item, 29% of the feasible solutions are in the 54% subsample. We can interpret the most right leaf node as, when not packing items five, six and seven, we increased the chance of the solution being feasible to 80%. The training data contains 70 observations, of which 11 observations were modelled (see Fig. 5).

Let us compare the decision tree with the optimal solution displayed in Table 6.

Table 6 tells us that decision variables x_1 and x_6 must be one for an optimal solution. By definition, the optimal solution is feasible. The feasibility tree (Fig. 5) agrees with this knowledge. We did not exclude these variables. The balanced training accuracy of 81.6% looks promising and induces some trust. The balanced test accuracy of this illustrative example is 79.6%, enhancing the trust. However, these values may be misleading since we are dealing with a single test data sample, which is small (30 observations). That means different data partitions lead to other accuracies. A more objective approach is obtained using repeated cross-validation (CV). Hence, we use a 5-fold CV with ten repetitions (i.e. 50 sample splits), learning classification trees with complexity parameters (CP) between 0.01 and 0.15, and investigated several quality measures, giving us a balanced accuracy $B = 68.7\%$, $F_1 = .448$ and $\kappa = .377$(Cohen Kappa) on average (based on the 50 data splits and the chosen (CP). The final models were all identical and identical to the tree structure displayed previously. This enhances the trust in the initial and current models. This, on its own, demonstrates that essential characteristics about the feasible region of an optimisation challenge can be derived using machine learning. The decision tree is interpretable and agrees with the intuitive knowledge that packing heavy items leaves fewer options for packing lighter items. Yet, it does not exclude packing the heavy item four (x_6) required for the optimal solution.

Synthetic oversampling: We could attempt to oversample the minority class (feasible) solutions to obtain a better prediction model. Interestingly, popular software packages such as SMOTE and ROSE do not offer methods for oversampling binary feature spaces. This is an interesting challenge on its own and can be addressed using local neighbourhood searches or other heuristics. Here, the immediate neighbourhood can be defined by flipping one bit (i.e. a changing one decision). Additionally, the previously learned feasibility classifier can help derive feasible solutions. This allows a targeted approach for under and oversampling. Later, when using XAI's LIME, the neighbourhood proves to be an important aspect. Admittedly, feasible solutions can be easily obtained in simple examples such as the Knapsack problem with few items (e.g. $n \leq 20 \Rightarrow 2^n \leq 1,048,576$ solutions). For larger instances, algorithmic solutions that consider runtime and memory are needed. At this

Table 6 Knapsack instance in Excel with problem and solution visualisation using ordered weights

	max	
Total Value		**15**
Packed weight		9
Weight limit	≤	9

item	*1*	*2*	*7*	*5*	*3*	*4*	*6*
x	1	2	3	4	5	6	7
Pack	b **1** b	0 b	0 b	0 b	0 b	**1** b	0
Weight	2	3	4	5	6	7	9
Value	6	5	3	6	8	9	7

point, the objective function could guide the selection of higher-quality solutions. Overall, synthetic oversampling helps us to generate more feasible solutions.

4.4 Interpretable machine learning solutions

Interpretable machine learning (IML) models are part of explainable AI. Classic examples of IML are decision tree models, linear regression models and logistic regression models. In machine learning, a model is a solution from a learning algorithm for a given target (output) and input features. Now, we will derive a machine learning model, predict its best solution value and compare it with the actual OR optimum.

A regression tree (Fig. 6) is learned using a standard ML algorithm such as the one from the rpart library (Therneau & Atkinson, 2023). However, the input data is restricted to feasible solutions only (see Sect. 4.3). A technical note: The algorithm's control parameters allow splitting nodes with two observations, the tree can grow to a maximum depth, and leaf nodes can contain one observation.

The tree gives us insights into the constellation of decision variables to the solution value. First, we observe that the decision tree provides the optimal solution (highlighted in red). It is easy to interpret by reading pack items one and six, resulting in the leaf node with solution value 15. Hence, we have successfully demonstrated that machine learning can be used to solve combinatorial optimisation challenges. The root node shows that the average solution value is $\bar{y} = 9.3$ given $n = 15$ feasible solutions. In this example, we created a decision tree that determines values for all solutions. In cases with many items, having several items in the leaf nodes will be preferable. This enables the system to communicate the average solution for a limited number of decision variables. However, expanding the tree for the optimal solution might be necessary.

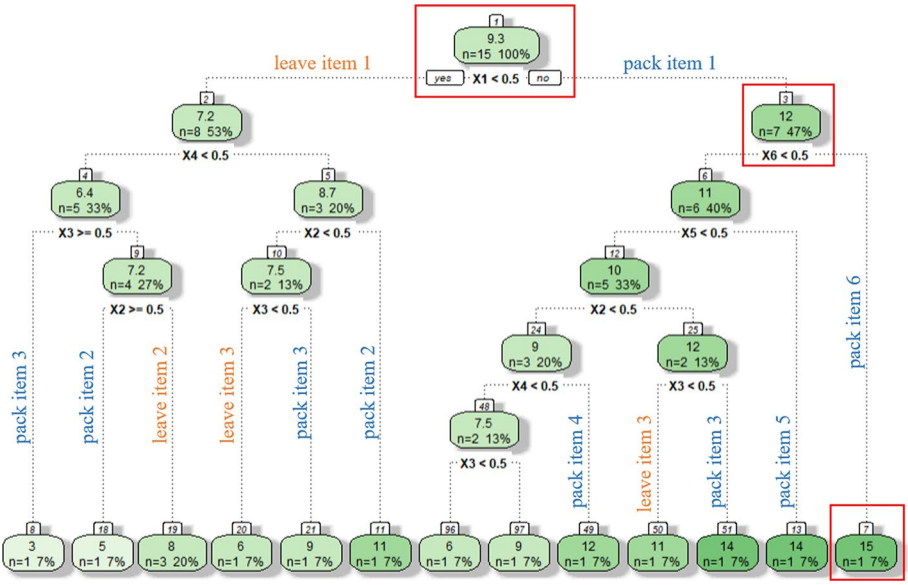

Fig. 6 Regression tree predicting solution value

4.5 Summary and implications

In this section, we derived two interpretable decision trees to explain the feasibility and solution. An *interpretable technique/model/result* is "easy" to explain to people (even to those without technical training) without ambiguity. Moreover, an *interpretable system* is one in which a user can, not only see but also study and understand how inputs are mapped to outputs. The first DT can be used to interpret feasible and infeasible solutions. The second DT is used to interpret the solution value. Generally, interpretable AI/ML may sacrifice accuracy in favour of interpretability.

This has *theoretical implications*. For instance, the OR-ML methodology introduced here outlines a novel application of intrinsic XAI in combinatorial optimisation (CO), providing a basis for future studies that explore machine learning techniques to enhance interpretability and explainability in CO. This offers a methodological contribution by structuring the steps needed to link optimisation problems with XAI, contributing to the theory of explainability in optimisation contexts, which can advance *transparency frameworks in OR*. Previously, we argued that feasibility classifications can be used to support explanations. However, learning machine learning models to classify feasible sub-spaces within combinatorial optimisation can also *enhance our understanding of search spaces*. This provides insights into how classification models capture the optimisation's constraints. This could potentially benefit combinatorial optimisation, especially those based on meta-heuristics (e.g., genetic algorithm). However, this also opens up the observed issue that a region classified as infeasible can contain feasible or even optimal solutions. Future studies could explore this in more detail. This work demonstrates the synergy between machine learning (focusing on predictions and classifications) and combinatorial optimisation. This initiated the acquisition of theoretical knowledge about how these fields interact, suggesting novel approaches for solving CO problems while offering transparency.

A *practical implication* of the OR-ML steps is the capability of *real-time decision-making*. The trained machine learning models can predict solution feasibility and values faster than CO algorithms. That means informed real-time decision-making for complex optimisation problems is possible. This is particularly valuable in environments such as traffic management (e.g., re-routing of buses, road closures, etc.) and emergency response planning. Rapid decision-making is required in such situations, and the ability to justify these decisions is critical for operational transparency and accountability. The importance of transparency warrants another practical implication: *transparent decision support systems* (TDSS). The provided OR-XAI procedure may assist in the creation of such systems for real-world applications requiring CO. These applications may deal with resource allocations, lot-sizing and supply chains. This can support practitioners to understand the reason behind decisions, which in turn can enhance trust and encourages the use of responsible AI.

5 Discussion: Post-hoc XAI in the context of combinatorial optimisation

In this Section, we will focus on local interpretable model-agnostic explanations (LIME) and human-friendly explanations, with their application to the knapsack example.

5.1 LIME

Local interpretable model-agnostic explanations (LIME) are popular. The explanation $\xi(x)$ for instance x is obtained by:

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}}\ \mathcal{L}(f, g, \pi_x) + \Omega(g),$$

where f is the predictor ($f : \mathbb{R}^m \to \mathbb{R}$, m number of features), g an interpretable model (from a class G of such models), $\pi_x(z)$ a proximity measure between x and z and $\Omega(g)$ is the complexity of g. LIME (Ribeiro et al., 2016) - in its basic form, attempts to find a simple "explanation" for a single instance. It is in the form of a linear regression or decision tree with a few features. LIME uses the classifier on a sample and weighs the output by its proximity. In its original version (Ribeiro et al, 2016, Alg. 1, Sparse linear explanation using LIME) an interpretable version of x is used, abbreviated by x'. Ribeiro et al (2016, S3.1) intention seems to have been $g : \mathbb{B}^{m'} \to \mathbb{B}$, but in later implementations, it appears that a selection of features was preferred - driven by $\Omega(g)$. LIME is model agnostic, which allows us to explain black-box models such as random forests or deep neural networks. Still, it also works on already interpretable models like the ones we derived earlier in this text. We apply LIME to the feasibility tree as an example. We need to specify the instance for which we want an explanation. An instance is a single observation (record). This should not be confused with an OR test instance, where we refer to the entire problem data. We want an explanation, for instance, ten $x_{10} = [1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0]$. Figure 7 displays such an explanation.

As we can see it agrees with the previously derived decision tree's explanation. Instance 10 is feasible because the features x_5, x_6 and x_7 values are zero. The underlying method is k-LASSO with the equation $\hat{y}_{10} = -.271 + .340[x_7 = 0] + .242[x_6 = 0] + .19[x_5 = 0] = .502$. We have not gained any new insights because we used an interpretable model previously. However, if it had been a black-box model, this would have been an insightful explanation. Similarly, we can obtain an explanation for the optimal case. It is interesting to observe that the neighbourhood size and its sampling size are important. A larger size can introduce a negative support of $x4 = 0$ and $x2 = 0$, reducing the explanation fit to 0.65. Generally, the definition of the neighbourhood for LIME is a debated topic in machine learning.

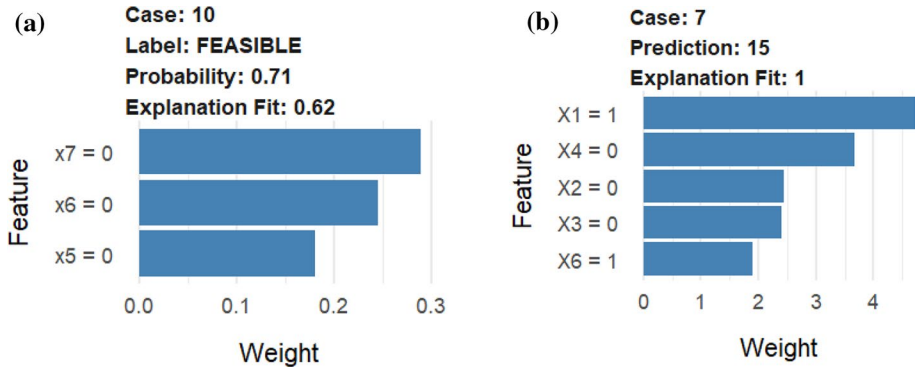

Fig. 7 LIME explanation for (**a**) feasibility of instance ten and (**b**) optimality of instance seven

Overall, we showed that LIME delivers an interpretable model with reduced model complexity for specified instances.

In the next section, we return to (probably) the original intention of LIME to generate locally interpretable human-readable explanations.

5.2 Transparent human-friendly explanations

Here, we provide a simple procedure to translate LIME into natural language for the audience as one objective of this work. Humans prefer short explanations, such as one or two sentences (Molnar, 2023a). A human-friendly explanation can be contrasting. That is, the current and its "inverse" are considered. Humans tend to focus on abnormal causes (Kahneman & Tversky, 1981), which pertains to the domain of counterfactual explanations. Another important aspect is the "social" interaction between the explainer and the explainee. In contrast, it is not human-friendly if an explanation requires all features. In Sect. 3.1 we provided more theoretical background about explanations and pointed to literature such as Miller (2019).

Let us return to the procedure that translates LIME into natural language. At this point, it is important to reconsider mapping the feature names (decision variables) to human-understandable terms. In the Knapsack problem, this can be as simple as naming the decision variable x as item. If there are numerous features, with some having similar characteristics, then a feature reduction via aggregation may be needed. In the Knapsack example, this may happen due to their weight, value, or benefit ("valuability"). For instance, in our "proof-of-concept" Knapsack example with $n = 7$ items, decision variables x_1,x_2 and x_3 could be represented as a single feature named "light items". Even if the features are not aggregated, the eventual explanations may use such a coding approach. Table 7 shows human-friendly terms for the Knapsack problem.

In the Knapsack problem, the variables weight and value are used, and benefit (valuability) is used indirectly. Their domain ranges are mapped to readable expressions, and their general form is shown in Table 7's neutral column.

Furthermore, the domains of each feature need to be reconsidered. In our Knapsack example, the 0-1 domain can be recoded in "leave" and "pack". Additionally, the parameter terms should be taken into account. This concept can be similar to or the same as what is used for feature naming. Generally, they will differ. To transform continuous numeric monotone-increasing values into predefined ordered language terms, we could use the mapping function shown in Code 1, assuming the language terms map to a uniformly distributed domain.

Table 7 Human-friendly terms for the Knapsack problem

Valuable	Weight	Value	Neutral
Most valuable	Very heavy	Very high value	Very high
Very valuable	Heavy	High value	High
Valuable	Normal weight	Normal value	Medium
Less valuable	Light	Low value	Low
–	Very light	Very low value	Very low

Table 8 Language terms mapped to ranges (a) weights; (b) benefits

(a) Weight terms	Lower	Upper	Items	(b) Benefit terms	Lower	Upper
Very light	-Inf	3.4	1, 2	Less valuable	-Inf	1.3125
Light	3.4	4.8	3	Valuable	1.3125	1.8750
Normal weight	4.8	6.2	4, 5	Very valuable	1.8750	2.4375
Heavy	6.2	7.6	6	Most valuable	2.4375	Inf
Very heavy	7.6	Inf	7			

Table 9 Human-readable terms for the Knapsack problem

Item	Weight	Value	Benefit	Weight term	Value term	Benefit term
1	2	6	3.00	Very light	Normal value	Most valuable
2	3	5	1.67	Very light	Low value	Valuable
3	4	3	0.75	Light	Very low value	Less valuable
4	5	6	1.20	Normal weight	Normal value	Less valuable
5	6	8	1.33	Normal weight	Very high value	Valuable
6	7	9	1.29	Heavy	Very high value	Less valuable
7	9	7	0.78	Very heavy	High value	Less valuable

Code 1 Map uniformly distributed terms to ordered terms (R-code).

```
getUniformRanges <- function(values, terms){
  nb  <- length(terms)                  # number of ranges
  s   <- (max(values) - min(values))/nb # step size
  as  <- agg.sum(min(values),s, nb)     # aggregated sum
  low <- as[1:nb];     low[1] <- -Inf   # lower range value
  up  <- as[2:(nb+1)]; up[nb] <-  Inf   # upper range value
  df <- data.frame(terms=terms,low=low, up=up) # map
  return(df)
}
agg.sum <- function(start=0, increment=1, nb=1){
  s    = rep(0,nb+1);  s[1] = start
  if (nb > 1) {for (k in 2:(nb+1)){s[k] <- s[k-1]+increment}}
  return(s)
}
```

Table 8 shows the output of Code 1 for the weight and benefit terms.

For instance, the normal weight term is mapped to the right-open interval $[4.8, 6.2)$ and applies to items four and five. After recoding, each item has the following human-friendly terms associated with it (Table 9).

The concept of mapping numeric values to human-readable terms can be easily extended to accommodate other distributions, such as the normal, exponential or Weibull distribution. The target can even be recoded, and its domain can be adapted as before. Another interesting approach to providing human-friendly explanations is discussed in Hagras (2018), where they propose the use of fuzzy logic systems (FLS). Their work confirms the fuzziness of human-language terms such as heavy and light and how well humans can deal with them. The FLS may utilise our procedure to generate human-readable systems or vice versa. These approaches are related to a rudimentary concept-based perspective. Kazhdan et al. (2021) and Ghorbani et al. (2019) discuss concept-based approaches in more detail. The search for human-readable explanations is not new. Confalonieri et al. (2020) provide a historical perspective of explainable AI, which reminds the reader of Expert Systems, Recommender Systems and Neuro-Symbolic Learning and Reasoning, besides the popular Machine

Learning area using explanations and XAI. Furthermore, their work details the explanation aspects for each area. For instance, for Expert Systems, they show how explanations can be derived as a problem-solving activity. Unfortunately, in the Machine learning domain, they stop at LIME as the "human readable" explanation. Yet, it may be worthwhile exploring the approaches from the other areas mentioned to derive a unified way to derive explanations. Generative AI is an exciting and popular domain. Schneider (2024) explores its relationship to XAI and provides a research agenda. The work captures some of the above-mentioned concepts and many more, basically indicating that the here proposed human-friendly explanation is just the tip of the iceberg.

Now, let us return to our proposed explanation system. There are several general, concrete approaches that should be considered when creating human-friendly explanations. The first general idea is that features with common characteristics can be identified via clustering. In our case, this was done indirectly by introducing a mapping from numerical to language terms (clusters). Obviously, other clustering techniques, such as k-means, could have been used, leading to fewer decision variables. In our example, for the feasibility analysis, we can reduce the dimensionality from seven to three weight decisions (light, normal, heavy). This becomes more impressive with the knapsacks of thousands of decision variables. The reduction of decision variables could have also used the value and benefit. The second general idea is to reduce the parameter space. This dimensionality and parameter space reduction has also been applied to the study of explainability in clustering algorithms (Rodriguez et al., 2019; Bandyapadhyay et al., 2023). For instance, the proposed mapping can be used to discretise the continuous range domains.

We can now return to our case study (scenario) and derive explanations for case 7 using the above mappings and natural language models, which can be derived from natural language processing (NLP). Let us begin by explaining the optimal solution using the parameter space reduction and textual representation:

- "*Item one was packed because it is most valuable*",
- "*Item six was packed because it has a very high value*",
- "*Item two was not packed because it is not valuable*", and
- "*The packed items provide the highest value*".

Returning to the SAGE framework, we can see that the logistics manager (audience) received a textual explanation (goal) of why this is an optimal selection of the packed items. For the optimal solution (case 7), explaining the varying characteristics by using the decision variable reduction approach is challenging. However, we can derive an explanation for the feasibility by introducing weight decision variables. For instance:

- "*Case 10 is feasible because no heavy items were packed*", and
- "*Case 10 is feasible because light items were packed*".

Here, we are not concerned with omitting some information. Instead, we promote it to favour simplicity because humans prefer short explanations (one or two sentences). It is worth noting that we are touching upon counterfactual explanations here (see Sect. 3.2). Once again, the SAGE's goal is met by explaining why a solution is feasible. Another expected expla-

nation concerning the setting of the SAGE framework is regarding the components of the system (Mill et al., 2024a, b).

- "*The weight of the packed items does not exceed the knapsack's limit*".

5.3 Implications

Generally, with respect to explainability, the combination of machine learning and operational research optimisation offers several advantages: (1) Solutions can be explained in a meaningful way to the audience providing the appropriate context in an accessible human-friendly way. (2) Results can be effectively contrasted. That means the characteristics of good and bad solutions can be learned. Additionally, feasible and infeasible solution properties can be differentiated. (3) Counterfactual explanations can be used in the context of the operations, describing the effects of using different inputs and values and their effects on the solution values.

The *theoretical implications* of this research include the development of a framework that integrates XAI into combinatorial optimisations in an operational research context. This is achieved by utilising XAI methods such as LIME and SHAP, building on SAGE and human-friendly explanations to provide transparency of OR solutions for decision-makers. Thus, our methodology is paving the way for *new transparency frameworks*. Furthermore, we *advance post-hoc XAI methods* by demonstrating their applicability to operational research scenarios. The work challenges the limitation of XAI methods from traditional prediction and classification tasks to being usable for explaining combinatorial optimisation search spaces and solutions.

The *practical implications* of this work include *enhanced decision-making* in industries such as manufacturing, finance and logistics. For instance, in resource allocation in manufacturing, using XAI for combinatorial optimisations helps decision-makers better understand how factors like value, volume and weight influence the operations. Another implication is the *improved trust in AI and OR systems* utilising our work by offering human-readable explanations. This can benefit non-experts of AI and OR, such as stakeholders in government and managers, allowing them to sense check solutions from these decision support systems quickly. This is especially important for sectors where transparency is legally or ethically required, e.g. in healthcare and critical public energy and transport infrastructure.

In summary, operational research optimisation solutions can be translated into transparent, human-friendly explanations. Overall, the arguments provided are conclusive, sustainable, and transparent, addressing the ethical dimension of the SAGE framework. This allows us in this golden age of artificial intelligence to make responsible decisions that are easy to understand and validate by the targeted audience, inducing trust in operational research optimisations.

6 Conclusion

This work emphasised the significance of transparency and responsible decision-making, particularly by applying machine learning (ML) to operational research (OR) optimisations with explainable AI (XAI). Using the SAGE framework, we highlighted the importance of

generating human-friendly explanations tailored to the audience's context, thereby increasing trust and transparency in OR solutions. We addressed the key objectives by:

- Contextualising transparency for OR applications: Using the SAGE framework, we guided the generation of XAI in generating transparency in OR, ensuring that the explanations provided are audience-specific and enhance trust.
- Demonstrating interpretable ML for combinatorial optimisations: Through the Knapsack problem example, we showed that interpretable ML models could solve combinatorial optimisation challenges, providing transparent and interpretable solutions. This intrinsic XAI provides transparency by featuring interpretable models. Additionally, they offer valuable insights into the solution space, detecting feature relationships and concepts that facilitate the generation of transparent solutions.
- Applying post-hoc XAI methods: We illustrated how post-hoc XAI methods, like LIME, can be applied to ML solutions for OR optimisations, visualising and explaining the solutions.
- Offering human-readable explanations: We proposed practical guidelines to improve the human readability of XAI methods in the context of OR challenges, guided by the SAGE framework.

Limitations: While our findings showed that interpretable ML techniques can estimate the feasibility space and find the optimal solution for a Knapsack problem instance, there are some limitations to our study. Although we provide a general methodology, it was only tested on one CO challenge. That means (i) the *generalisability and applicability* to other operational research optimisation problems, such as vehicle routing, dynamic lot sizing and facility location problems, need to be further investigated. Furthermore, our approach and analysis were methodological, preparing the way for (ii) a *theoretical study*. Many questions with a theoretical direction arose during this study, including: Are the information gain and minimum-squared error appropriate loss functions? What are the mathematical differences between the objective and loss functions? What are the algorithmic similarities between decision trees and branch-and-bound algorithms? What is the difference between data-driven function estimation of the feasible region and solution-specific search? Our approach still shares the limitations associated with using ML to solve combinatorial optimisation problems, namely (iii) *reduced accuracy*, as ML methods often fail to capture subtle problem-specific nuances, and (iv) the challenge of incorporating complex real-world constraints into ML algorithms. This may constitute challenges when exploring complex OR scenarios.

Nevertheless, this study demonstrated that combinatorial optimisations could benefit from ML because of the enhanced transparency of solutions achieved by using XAI enhanced with human-friendly concepts. This research opens up opportunities for increased transparency in operational research across various sectors.

Future research directions include:

- Exploring multiple XAI methods: Applying various XAI methods to other classic OR problems.
- Utilising diverse ML techniques: Different ML techniques, such as logistic regressions, random forests, or neural networks, can be used to learn the feasible region of the search

space.

- Investigating metaheuristics: Examining metaheuristics in OR problems from the perspectives of interoperability, explainability, and transparency, for both traditional and parallel (Amirghasemi, 2022) approaches.
- Exploring generative AI and large language models (LLMs): We are investigating the potential of generative AI and LLMs, trained on data from the OR problem search space, to provide human-friendly explanations. These future directions will further advance transparency by means of interpretability and explainability in OR optimisations, leveraging the benefits of ML and XAI.

Declarations

Conflict of interest The authors have no conflict of interest to disclose. All co-authors have seen and agree with the contents of the manuscript and there is no financial interest to report.

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